

1 Detailed properties of the short chain

1.1 Stabilization diagnostics from the reconstructed stationary density

We next use the normalized five-dimensional stationary density from Section 1 to characterize the nonequilibrium stabilization mechanism. The analysis is performed for $(T_1, T_3) = (2, 8)$ and is restricted to the validation-defined sampled-support mask, which contains 99.420% of the held-out stationary probability and 99.422% of the reconstructed probability. All density-derived results are compared with independent held-out controlled-integrator trajectories; the equal-temperature case $(5, 5)$ calibrates the numerical floor. Uncertainties are obtained by resampling complete trajectories rather than individual time samples.

Exchange-symmetry breaking. Let $\sigma(I_1, I_2, I_3, \theta_1, \theta_3) = (I_3, I_2, I_1, \theta_3, \theta_1)$ and define

$$A_\sigma(x) = \frac{|\rho_{ss}(x) - \rho_{ss}(\sigma x)|}{\rho_{ss}(x) + \rho_{ss}(\sigma x)}. \quad (1)$$

For the driven density, the mean, median, and 90th percentile of A_σ are 0.1796, 0.1153, and 0.4471, respectively; the three training seeds give angular total-variation asymmetries in the narrow range 0.1141–0.1157. The ensemble value, 0.1149, agrees with the independent trajectory value 0.1159. At equilibrium the exact-density pointwise asymmetry is zero to floating-point precision, whereas the finite 60×60 histogram produces a measured TV floor 0.0287. The driven-to-equilibrium point-estimate ratio is therefore 4.00. The deliberately conservative predeclared gate, however, compares the driven lower bootstrap limit with three times the equilibrium upper resampling floor; that ratio is only 2.44. We consequently report robust descriptive symmetry breaking but do not label the strict three-times-floor gate as passed.

Energy-dependent phase locking. We divide the sampled support into energy terciles fixed on the validation set and compare the conditional angle distributions with the high-energy branch

$$\theta_0 = \pi - \arcsin \left[\frac{\sqrt{4\gamma^2 + 3} - \gamma}{2(1 + \gamma^2)} \right] = 2.1912, \quad (2)$$

using a fixed angular window of half-width $\pi/6$. The trajectory probability $P(|\theta_1 - \theta_0| < \pi/6)$ increases from 0.2041 in the low-energy tercile to 0.2505 in the high-energy tercile. The reconstructed values are 0.2043 and 0.2447, compared with 0.2006 and 0.2230 for the legacy density. The analogous high-energy probability for $|\theta_3 + 2\pi/3| < \pi/6$ is 0.2189 directly and 0.2182 from the new density. Thus the earlier phase-locking mechanism persists and is reproduced more accurately by the normalized density.

The comparison is not perfect: 27 of 30 predeclared density/trajectory confidence intervals overlap. The three failures are the low-energy θ_3 circular mean (difference 0.0351 rad), the middle-energy θ_1 circular mean (difference -0.0300 rad), and the high-energy θ_1 locking probability (difference -0.00578). We retain these small but statistically resolved discrepancies rather than selecting only the agreeing observables.

Stationary current balance. For the unforced middle mode, define $J_{12} = I_1 I_2 \sin \theta_1$ and $J_{23} = I_2 I_3 \sin \theta_3$. Stationarity requires the global identity

$$\langle J_{12} + J_{23} \rangle_{ss} = 0. \quad (3)$$

On the complete held-out driven trajectories we obtain $\langle J_{12} \rangle = 0.232750$, $\langle J_{23} \rangle = -0.232733$, and $\langle J_{12} + J_{23} \rangle = 1.70 \times 10^{-5}$ with a 95% interval $[-2.70, 3.17] \times 10^{-4}$. The global current-balance identity therefore passes directly.

Density expectations cannot be extended outside the validated support. On the common support mask, the direct current sum is 0.001926 and the new density gives -0.003847 ; their difference is -2.31 bootstrap standard errors. The legacy density gives 0.12375 with a much larger uncertainty and only 3.37% importance-sampling effective sample fraction. Conditioning on the support mask destroys the exact zero identity, so the masked comparison is a reconstruction test rather than a second stationarity test. That reconstruction test remains a declared failure, even though the new density is substantially closer than the legacy model.

Taken together, the controlled data support the qualitative stabilization picture: the drive breaks endpoint-exchange symmetry, phase locking strengthens with energy, and the two bond currents balance globally. Quantitative claims derived from the density are restricted to the sampled-support region and are reported together with the three phase discrepancies and the masked-current mismatch.

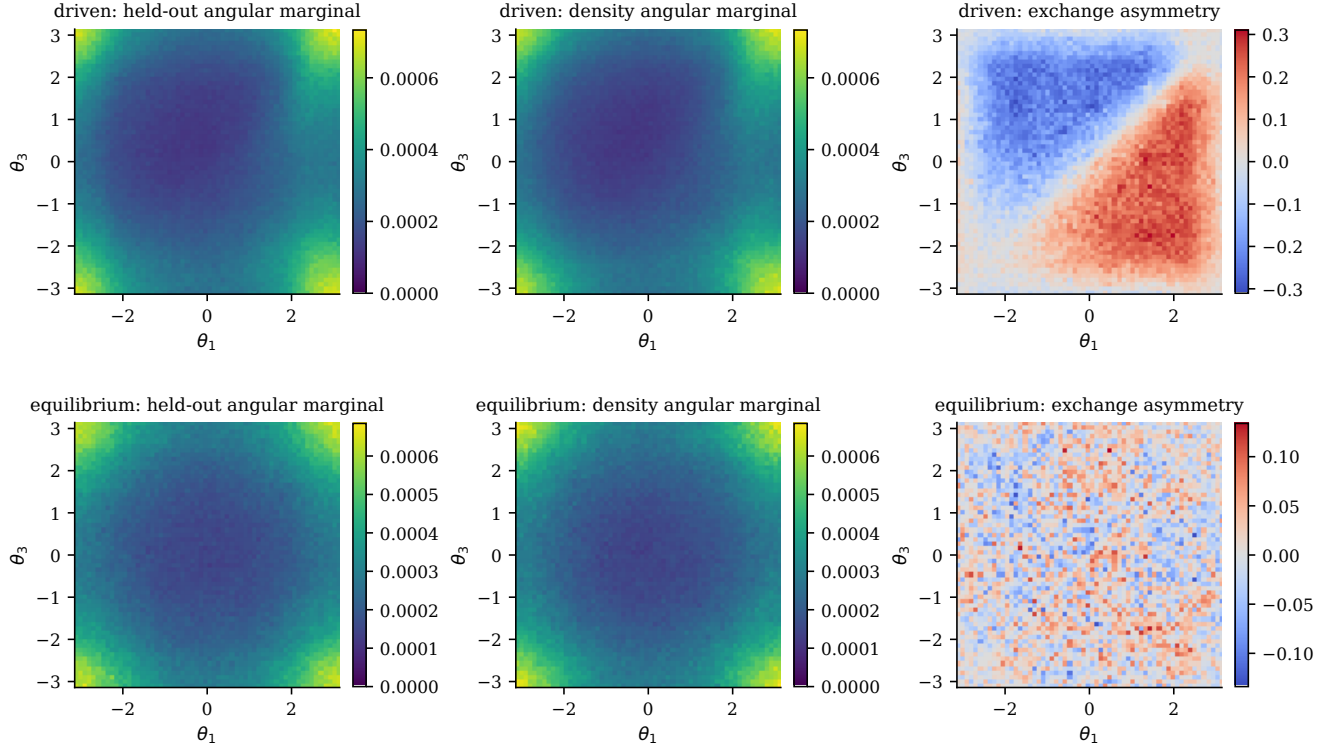


Figure 1: Endpoint-exchange symmetry breaking in the angular stationary marginal. The validated density follows the held-out trajectory marginal, while the equilibrium panel calibrates the finite-histogram floor. All density evaluations are confined to the sampled-support mask.

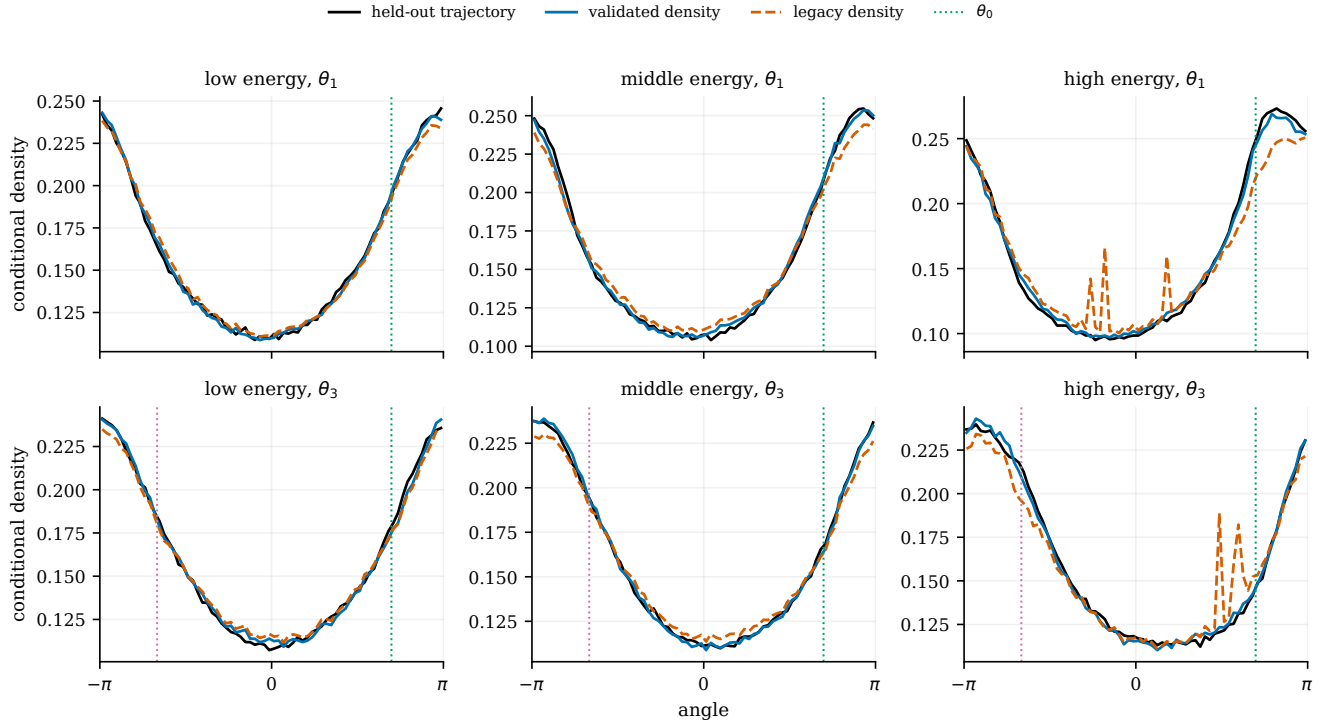


Figure 2: Energy-conditioned angular densities from held-out trajectories, the validated stationary density, and the legacy density. Energy sectors, bins, and reference angles were fixed before test evaluation.

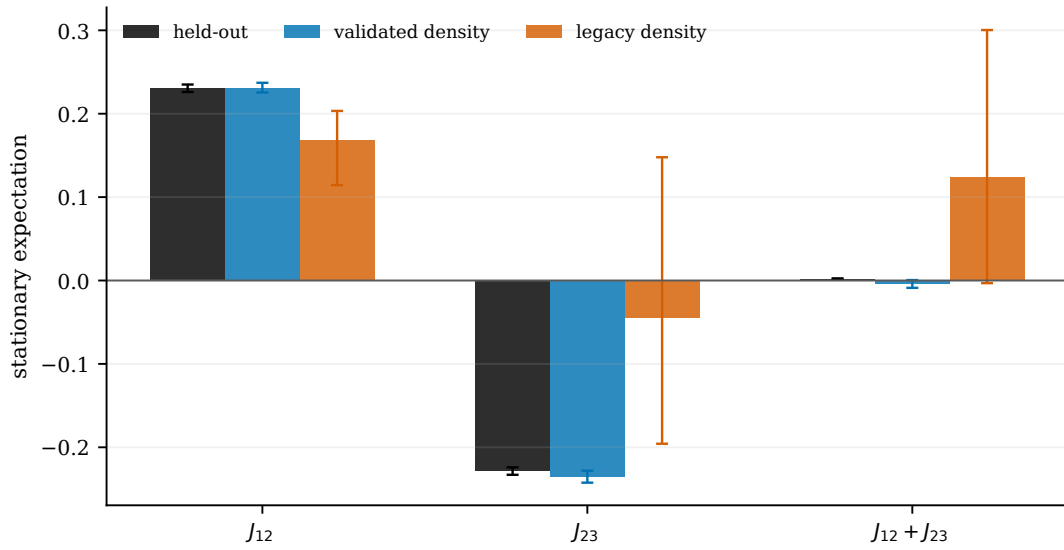


Figure 3: Current expectations on the validation-defined support mask. The validated density improves substantially over the legacy result, while the residual masked density/trajectory discrepancy remains explicitly visible.